



Food Security and Health Disparities in the US

Lanqi Huang



Introduction

Project Motivation

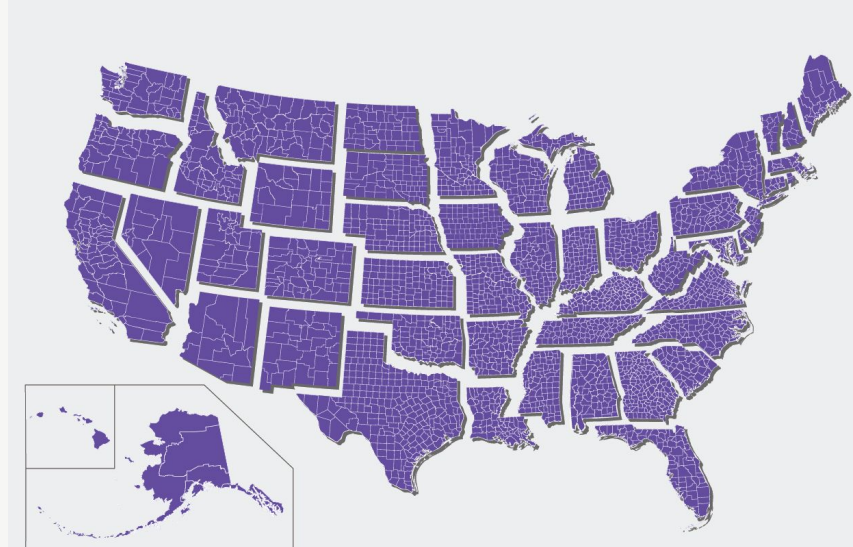
Food Environment → Health Disparities

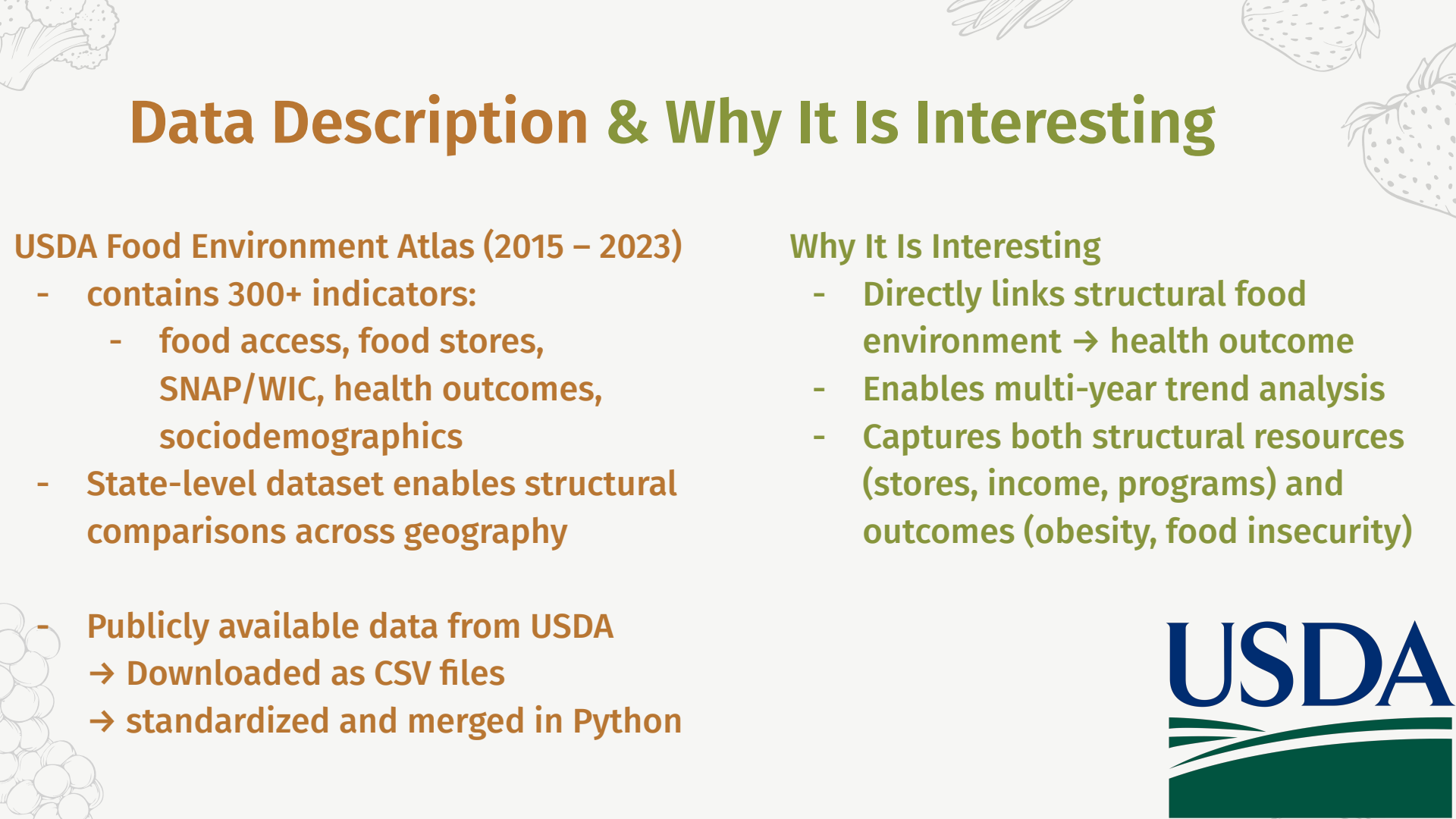
- Food access is a structural determinant of health
- Geographic inequalities shape nutrition & chronic disease outcomes
- Millions of U.S. households face barriers to obtaining healthy foods

Research Question

- Do food-insecure states show higher obesity and diabetes?
- How does food access change over time?
- What structural clusters can we identify across states?

Goal: Build an interactive analytical dashboard to explore structural food environments and health disparities across U.S. states





Data Description & Why It Is Interesting

USDA Food Environment Atlas (2015 – 2023)

- contains 300+ indicators:
 - food access, food stores, SNAP/WIC, health outcomes, sociodemographics
- State-level dataset enables structural comparisons across geography
- Publicly available data from USDA
 - Downloaded as CSV files
 - standardized and merged in Python

Why It Is Interesting

- Directly links structural food environment → health outcome
- Enables multi-year trend analysis
- Captures both structural resources (stores, income, programs) and outcomes (obesity, food insecurity)





FAIRness of Data

F A I R



Finable

clear variable codes &
metadata



Accessible

open CSV download, no
barriers



Interoperable

CSV/JSON formats
(minor naming
inconsistencies)



Reusable

CC0 public-domain
license

Needs Improvement

metadata depth &
collection details



Data Preparation & Processing

**Selected 15 Variables
across 4 domains:**

- Food access
- Socioeconomics
- Assistance
- Health



**Cleaned & standardized
variables (z-score)**

< 3 % missing →
state-mean
imputation



**Implemented API routes
for scatter, change,
cluster**



Interactive Dashboard

ACCESS TO HEALTHY FOOD



Dashboard Variable Selection

15 variables selected (2015–2023):

- Food access
 - (PCT_LACCESS_POP15, PCT_LACCESS_POP19)
- Obesity & diabetes
 - (PCT_OBESE_ADULTS22, PCT_DIABETES_ADULTS19)
- Socioeconomic indicators
 - Poverty & income(POVRATE21, MEDHHINC21)
- Grocery stores
 - (GROC16, GROC20)
- Food assistance
 - SNAP participation, WIC redemptions(PCT_SNAP22, PC_WIC_REDEMP22)
- Food insecurity (3-year average)
 - (FOODINSEC_21_23, VLFOODSEC_21_23)
- Metro/nonmetro (urban–rural)
 - (METRO23)



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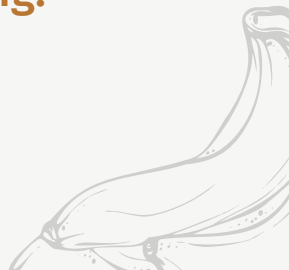
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Interactive Dashboard



Data Cleaning & Preprocessing

- Standardizing variable names
 - Converted all variable codes to uppercase.
 - Handling missing values
 - Missing values for one or two states (e.g., WIC redemptions missing data for DC).
 - Dropped states with missing values only for analyses involving that variable.
 - Summary statistics used pairwise deletion.
 - Cast data to numeric types
 - Some columns imported as strings due to commas or formatting.
 - Cleaned using `pd.to_numeric(errors='coerce')`.
 - Re-shaping
 - Combined variables into a “wide” format aligned by state.
 - Created Human-readable labels
- 

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Interactive Dashboard

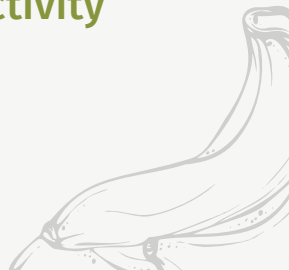


Server API and Web Interface

Features:

- Interactive state lookup
 - `/api/state/<state>`
- variable list
 - `/api/variables`
- Summary table button
 - `/api/stats`
- Heatmap generation
 - `/api/heatmap`
- Dynamic scatter plot generation
 - `/api/scatter/<var1>/<var2>`
- Customizable k for clustering
 - `/api/cluster/<k>`

Technical elements:

- Flask for API routing
 - Matplotlib for server-side plotting
 - JSON for data transfer
 - HTML/CSS/JS for interactivity
- 

Web Interface

Explore Food Security and Health Disparities by State

Enter a U.S. state name to see its food insecurity and obesity rates:

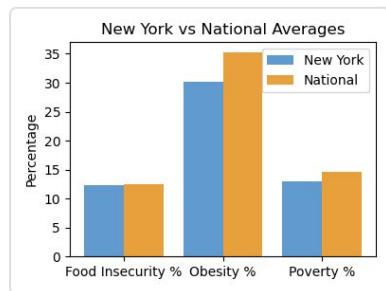
e.g., Alabama

Submit

[Go to Full Analysis Dashboard →](#)

Results for New York

In New York (NY), 12.3% of households are food insecure and 30.1% of adults are obese. Poverty rate is 13.0% (vs 14.6% nationally), median income \$67,844 (vs \$58,942 U.S. average).



[← Try another state](#)

[Go to Full Analysis Dashboard →](#)

Food Access & Equity Explorer

USDA Food Environment Atlas — Interactive Analysis

[← Back to Simple State Lookup](#)

1) Variables & Summary

Load variables

Summary stats

15 variables loaded.

2) Correlation heatmap (image)

Generate heatmap

3) Scatter plot

X variable (Variable_Code) Population, low access to st

Y variable (Variable_Code) Population, low access to st

Plot

4) State clustering (parameter: k)

k (2-12) 3

Run

5) State snapshot (your original endpoint)

State (full name or abbr) e.g., New York or NY

Fetch

Summary Statistics Dashboard

1) Variables & Summary

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Variable	Mean	SD	Min	Max	N
Population, low access to store (%), 2015	26.29	10.11	2.12	50.77	51
Population, low access to store (%), 2019	25.33	8.52	4.60	45.00	51
Adult diabetes rate, 2015	8.25	0.99	6.57	11.05	51
Adult diabetes rate, 2019	8.70	1.00	7.11	11.28	50
Adult obesity rate, 2017*	30.60	3.86	22.60	38.10	51
Adult obesity rate, 2022*	33.76	4.06	24.30	41.00	51
Poverty rate, 2021	13.82	3.39	8.51	22.25	51
Median household income, 2021	63360.81	11433.85	44196.09	91072.00	51
Grocery stores, 2016	32.80	42.75	3.08	186.00	51
Grocery stores, 2020	34.12	39.18	4.94	175.00	51
Metro/nonmetro counties, 2023	1.00		1.00	1.00	51
SNAP participants (% pop), 2022*	11.79	4.01	4.61	23.61	51
WIC redemptions per capita, 2022	10.56	3.46	3.06	20.95	49
Household food insecurity (% , three-year average), 2021-23*	11.70	2.30	7.40	18.90	51
Household very low food security (% , three-year average), 2021-23*	4.59	1.02	3.20	7.00	51

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Correlation Heatmap



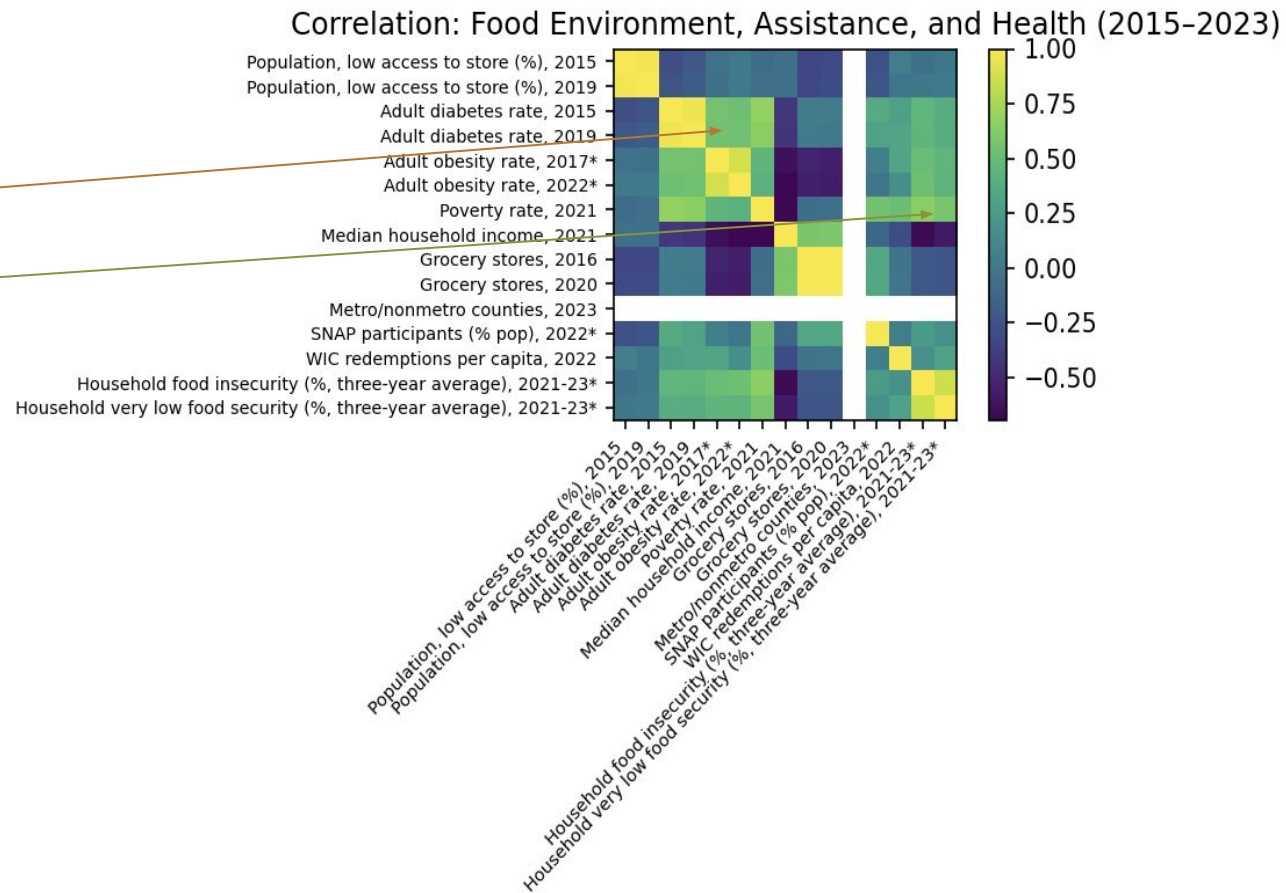
Obesity ↔ diabetes: strong block

Poverty ↔ food insecurity

SNAP/WIC moderate correlation with need

Grocery stores weak correlation with disease

Metro status not strongly predictive



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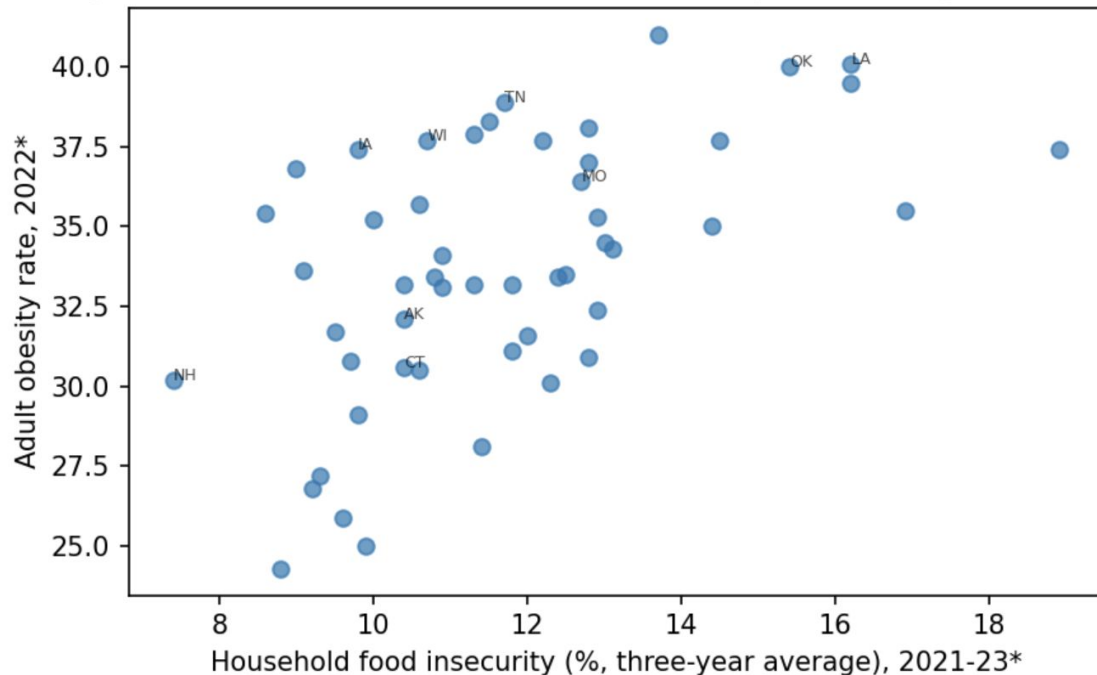
k (2-12) **Run**

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Scatterplot Example: Food Insecurity vs Obesity

obesity rate, 2022* vs Household food insecurity (% , three-year average)



Clear upward trend

Higher poverty → higher obesity prevalence

Outliers: LA, OK, WV (very high obesity)

Low-poverty, low-obesity: MA, CT, NH

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K-Means Clustering

4) State clustering (parameter: k)

k (2-12)

State	Name	Cluster
AK	Alaska	0
AL	Alabama	1
AR	Arkansas	1
AZ	Arizona	1
CA	California	2
CO	Colorado	0
DC	District of Columbia	2
DE	Delaware	1
FL	Florida	1
GA	Georgia	1
IA	Iowa	0
ID	Idaho	0
IL	Illinois	0
IN	Indiana	1
KS	Kansas	0
KY	Kentucky	1
LA	Louisiana	1
MA	Massachusetts	2
MD	Maryland	0
ME	Maine	0
MI	Michigan	1
MN	Minnesota	0
MO	Missouri	1
MS	Mississippi	1
MT	Montana	0
NC	North Carolina	1
ND	North Dakota	0
NE	Nebraska	0
NH	New Hampshire	0
NM	New Mexico	1
NV	Nevada	0
NY	New York	2
OH	Ohio	1
OK	Oklahoma	1
OR	Oregon	0
PA	Pennsylvania	0
RI	Rhode Island	0
SC	South Carolina	1
SD	South Dakota	0
TN	Tennessee	1
TX	Texas	1
UT	Utah	0
VA	Virginia	0
VT	Vermont	0
WA	Washington	0
WI	Wisconsin	0
WV	West Virginia	1
WY	Wyoming	0

(k=3)

- High-income/low-obesity (Northeast, West)
- High-poverty/high-obesity (South, Midwest)
- Urban coastal (CA, NY, MA)

(k=7)

- High-obesity, high-poverty states
- High-income, low-obesity coastal states
- Rural low-store-access states
- Mixed middle-of-the-distribution states

4) State clustering (parameter: k)

k (2-12)

State	Name	Cluster
AK	Alaska	3
AL	Alabama	0
AR	Arkansas	0
AZ	Arizona	5
CA	California	2
CO	Colorado	4
DC	District of Columbia	2
DE	Delaware	5
FL	Florida	5
GA	Georgia	1
IA	Iowa	6
ID	Idaho	6
IL	Illinois	6
IN	Indiana	1
KS	Kansas	3
KY	Kentucky	0
LA	Louisiana	0
MA	Massachusetts	4
MD	Maryland	5
ME	Maine	6
MI	Michigan	1
MN	Minnesota	6
MO	Missouri	1
MS	Mississippi	0
MT	Montana	3
NC	North Carolina	1
ND	North Dakota	3
NE	Nebraska	3
NH	New Hampshire	4
NM	New Mexico	5
NV	Nevada	4
NY	New York	2
OH	Ohio	1
OK	Oklahoma	0
OR	Oregon	1
PA	Pennsylvania	6
RI	Rhode Island	4
SC	South Carolina	0
SD	South Dakota	3
TN	Tennessee	1
TX	Texas	1
UT	Utah	4
VA	Virginia	1
VT	Vermont	6
WA	Washington	4
WI	Wisconsin	6
WV	West Virginia	0
WY	Wyoming	6

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State (full name or abbr)

e.g., New York or NY

Fetch

State Snapshot API Endpoint

5) State snapshot (your original endpoint)

State (full name or abbr)

Fetch

```
{
  "FOODINSEC_21_23": 12.299999999999995,
  "GROC16": 168.67741935483872,
  "GROC20": 143.34426229508196,
  "MEDHHINC21": 67843.80645161291,
  "METRO23": 1,
  "PCT_DIABETES_ADULTS15": 7.92741935483871,
  "PCT_DIABETES_ADULTS19": 8.982258064516131,
  "PCT_LACCESS_POP15": 16.365634754967743,
  "PCT_LACCESS_POP19": 16.11209273087097,
  "PCT_OBESE_ADULTS17": 25.700000000000003,
  "PCT_OBESE_ADULTS22": 30.099999999999999,
  "PCT_SNAP22": 14.618345920000001,
  "PC_WIC_REDEMP22": 10.168887581053571,
  "POVRATE21": 12.964516129032258,
  "VLFOODSEC_21_23": 4.6999999999999998,
  "abbr": "NY",
  "state": "New York"
}
```

NY profile:

- Obesity: ~30%
- Poverty: ~12%
- Food insecurity ~12%
- High grocery store availability
- Median income ~\$67k

Summary Statistics & Key Findings

- Low food access: 26.3% (2015) → 25.3% (2019)
- Obesity: ↑ 30.6 → 33.8 % (2017–22)
- Diabetes: ↑ 8.3 → 8.7 %
- Poverty: 13.8 % (mean)
 - (range 8.5–22.3%)
- Income range \$44k – \$91k
- Food insecurity: 11.7 % Very low security: 4.6 %
- Food insecurity, poverty, and obesity cluster tightly
- Income is a strong protective factor
- Food assistance reliably follows need
- Store availability varies dramatically across states
- Rurality remains a major divider in access



Challenge

- Variable inconsistency across years
 - Multi-year alignment required extra cleaning
 - Heatmap labeling challenges
 - K-means sensitive to scaling
 - Flask debugging (404 from wrong endpoints)
 - Rendering large images on the web page
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Conclusion & Policy Implication

- Health inequity is structural → not individual
 - Invest in rural grocery and transport infrastructure
 - Integrate nutrition policy with economic development
 - Simplify SNAP/WIC eligibility & increase benefits
 - Continue publishing FAIR open data for accountability
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References

- Centers for Disease Control and Prevention. *Behavioral Risk Factor Surveillance System (BRFSS) Data, 2022*.
- Drewnowski, A., & Specter, S. (2004). Poverty and obesity: the role of energy density and energy costs. *American Journal of Clinical Nutrition*, 79(1), 6–16.
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Thank you!